Extraction of Waveguide Group Index via Unbalanced Mach-Zehnder Interferometers

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Abstract: This report outlines the design and simulation of unbalanced MZIs for group index extraction in 500 nm x 220 nm SOI strip waveguides. Using a multi-device strategy with path imbalances ranging from 50 to 200 µm, the design maps the interference pattern's Free Spectral Range to the waveguide's dispersive properties. FDE and circuit-level PDK simulations confirm a group index of ~4.06–4.24, demonstrating the impact of structural dispersion. The resulting MZI array provides a high-confidence diagnostic tool for calibrating fabricated silicon photonic circuits against theoretical models.

1. Introduction

The objective of this work is to design a set of unbalanced Mach-Zehnder Interferometers (MZIs) specifically optimized to extract the group index of a 500 nm x 220 nm SOI strip waveguide. By introducing a known physical path length difference (ΔL) between the two arms of the MZI, the resulting interference pattern creates a periodic transmission spectrum. The spacing between these spectral peaks, known as the Free Spectral Range (FSR), provides a direct mathematical link to the group index. This report outlines the underlying physics, the modal simulation of the waveguide, the strategic variation of design parameters to ensure measurement reliability, and the final verification of the design through circuit-level simulations using foundry-standard Process Design Kits (PDKs).

2. Theory

2.1 Physics of the Unbalanced Mach-Zehnder Interferometer (MZI)

An integrated Mach-Zehnder Interferometer (MZI) consists of an input optical splitter (typically a Y-junction or Multimode Interferometer), two waveguide arms, and an output combiner. In an unbalanced MZI, one arm is physically longer than the other by a length difference $\Delta L = L_1 - L_2$.

As light propagates through the device, it acquires a phase in each arm. Upon recombination at the output, the interference pattern depends on the phase difference ($\Delta\emptyset$) accumulated between the two paths. Assuming the waveguides are identical in cross-section (sharing the same effective index n_{eff}), the phase difference is given by:

$$\Delta\emptyset = (2\pi n_{eff}\, \Delta L)/\lambda$$

Where n_{eff} is the effective refractive index of the waveguide mode, ΔL is the physical path length difference, and λ is the operating wavelength in vacuum.

2.2 Free Spectral Range (FSR)

The interference at the output results in a sinusoidal transmission spectrum when scanned across a range of wavelengths. Constructive interference (transmission peaks) occurs when the phase difference is an integer multiple of 2π, while destructive interference (transmission nulls) occurs at odd multiples of π.

The Free Spectral Range (FSR) is defined as the wavelength spacing ($\Delta\lambda_{FSR}$) between two consecutive transmission peaks (or resonance notches) in the spectrum.

2.3 Derivation of the Group Index (n_g)

In dispersive media like silicon waveguides, the effective index n_{eff} is a function of wavelength. To accurately characterize light propagation, we must use the group index (n_g), which accounts for this dispersion.

To derive the relationship between FSR and n_g, we start with the condition for a transmission peak at a mode order m:

$$\Delta\emptyset = \frac{2\pi n_{eff}\Delta L}{\lambda} = 2\pi m$$

Differentiating the phase difference $\Delta\emptyset$ with respect to the wavelength λ:

$$\frac{d\emptyset}{d\lambda} = 2\pi\Delta L \frac{d}{d\lambda}\left(\frac{n_{eff}}{\lambda}\right)$$

Applying the quotient rule:

$$\frac{d\emptyset}{d\lambda} = 2\pi\Delta L\left[\frac{1}{\lambda}\frac{dn_{eff}}{d\lambda} - \frac{n_{eff}}{\lambda^2}\right] = -\frac{2\pi\Delta L}{\lambda^2}\left[n_{eff} - \lambda\frac{dn_{eff}}{d\lambda}\right]$$

The term in the brackets is the definition of the group index: $n_g = n_{eff} - \lambda\frac{dn_{eff}}{d\lambda}$. Therefore:

$$\frac{d\emptyset}{d\lambda} = \frac{2\pi\Delta L.n_g}{\lambda^2}$$

For a change between two consecutive peaks, the phase changes by $2\pi(|d\emptyset| \approx 2\pi)$ over a wavelength shift equal to the FSR ($|d\lambda| \approx \Delta\lambda_{FSR}$). Replacing these into the magnitude of the equation:

$$2\pi = \frac{2\pi\Delta L.n_g}{\lambda^2}\Delta\lambda_{FSR}$$

Solving for n_g, we arrive at the final expression:

$$n_g = \frac{\lambda^2}{\Delta\lambda_{FSR}\Delta L}$$

2.4 How to extract n_g

By measuring the distance between two adjacent peaks ($\Delta\lambda_{FSR}$) at a central wavelength λ, and knowing the designed physical path difference ΔL from the layout, n_g can be calculated.

3. Modeling and Simulation

3.1 Waveguide Geometry and Environment

The core of the MZI consists of a standard Silicon-on-Insulator (SOI) strip waveguide. The geometry is optimized for single-mode operation in the C-band. The design parameters are as follows:

Core Material: Silicon (S_i), $n \approx 3.47$ at 1550 nm.

Cladding: Silicon Dioxide (S_iO_2) for both the Under-Cladding and Top-Cladding, $n \approx 1.44$.

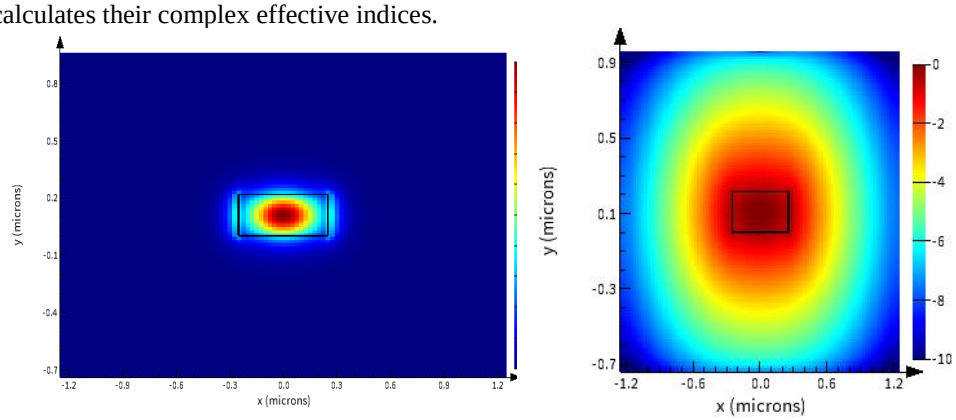
Width (w): 500 nm.

Height (h): 220 nm.

Polarization: Quasi-TE (Transverse Electric), which provides high confinement and is the standard for 500 nm strip geometries.

3.2 Modal Analysis

To determine the optical properties of the waveguide, a Finite Difference Eigenmode (FDE) solver (Lumerical MODE) was utilized. The simulation identifies the supported modes and calculates their complex effective indices.



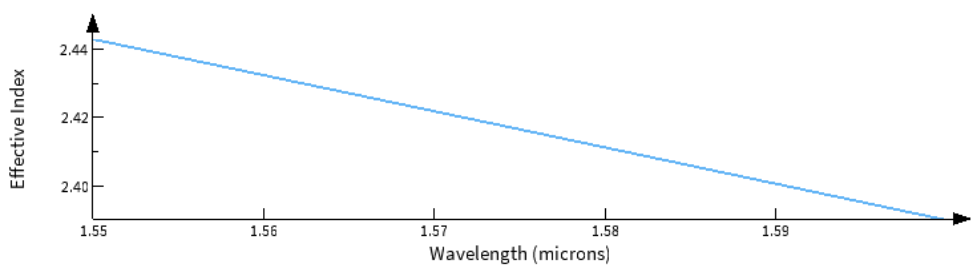
Mode Profile: The fundamental TE mode shows high intensity within the $500x220$ nm silicon core, with evanescent tails extending into the S_iO_2 cladding.

Power Confinement: The simulation indicates a high confinement factor (98%), ensuring that the majority of the optical power is contained within the silicon core, minimizing substrate leakage and sensitivity to top-cladding variations.

3.3 Index Simulations and Dispersion

The effective index (n_{eff}) and group index (n_g) were simulated over a wavelength range of 1500 nm to 1600 nm.

As shown in the generated plot, the n_{eff} decreases linearly as the wavelength increases, which is characteristic of normal chromatic dispersion.



While n_{eff} is approximately$\sim$2.44, the group index (n_g) is significantly higher $\sim$4.059. This discrepancy is due to the strong structural dispersion in high-index-contrast silicon waveguides.

3.4 Compact Model

To integrate the simulation results into a circuit-level model, the n_{eff} data was fitted to a 2nd-order polynomial as a function of wavelength (λ in μm). This allows for accurate interpolation of the phase during the extraction process.

The resulting compact model is:

$$n_{eff}(\lambda) = a\lambda^2 + b\lambda + c$$

Based on the simulation data for the $500x220$ nm geometry, the extracted coefficients are: $a = 0.125\ \mu m^{-2}$, $b = -1.482\ \mu m^{-1}$, $c = 4.451$.

3.5 Circuit Transfer Function

The behavior of the unbalanced MZI can be modeled by the power transmission (T) as a function of the phase difference between the arms. Neglecting propagation loss and assuming a 50:50 split ratio at the couplers, the mathematical expression for the normalized power transmission is:

$$T(\lambda) = \frac{1}{2}\left[1 + \cos\left(\frac{2\pi n_{eff}(\lambda)\Delta\lambda}{\lambda}\right)\right]$$

This sinusoidal transfer function is the basis for the experimental extraction. By measuring the distance between the peaks produced by this cosine term, we can reverse-calculate n_{eff} using the derived theory from Section 3.

4. Design Parameter Variations

4.1 Design Strategy

To ensure the robustness of the experimental data and to account for potential fabrication tolerances, a sweep of the path length difference (ΔL) is implemented. By varying ΔL, we can observe the interference pattern at different spectral resolutions. This multi-device approach serves two purposes:

Validation: It confirms that the extracted n_g is a property of the waveguide itself and is independent of the specific ΔL used.

Reliability: It ensures that at least one device will provide a clear, measurable Free Spectral Range (FSR) even if the actual group index deviates significantly from the simulated value.

4.2 Proposed Variations

The designs are optimized for the C-band (centered at $\lambda = 1550$ nm). The predicted FSR for each design was calculated using the previously derived formula:

$$\Delta\lambda_{FSR} = \frac{\lambda^2}{n_g.\Delta L}$$

For these calculations, an estimated group index of $n_g \approx 4.059 \approx 4.1$ (based on simulations section 2) was used.

Design	**ΔL (μm)**	**Predicted FSR (nm)**	**Logic**
MZI1	50	~11.7 nm	Safety: A wide FSR ensures multiple peaks are visible even with high group index or narrow OSA spans.

MZI2	100	~5.9 nm	Standard: Provides a balance between spectral resolution and ease of measurement.
MZI3	150	~3.9 nm	Resolution: Increases the number of periods within the C-band for better statistical averaging.
MZI4	200	~2.9 nm	Precision: High-density fringes allow for high-precision n_g extraction and dispersion analysis.

By fitting the measured FSR from all four designs against $1/\Delta L$, the group index n_g can be extracted from the slope of the resulting line, significantly reducing the impact of random measurement errors.

5. Circuit Simulation and Layout Verification

5.1 PDK and Workflow Integration

The simulation workflow involved a tight integration between physical layout and circuit behavior:

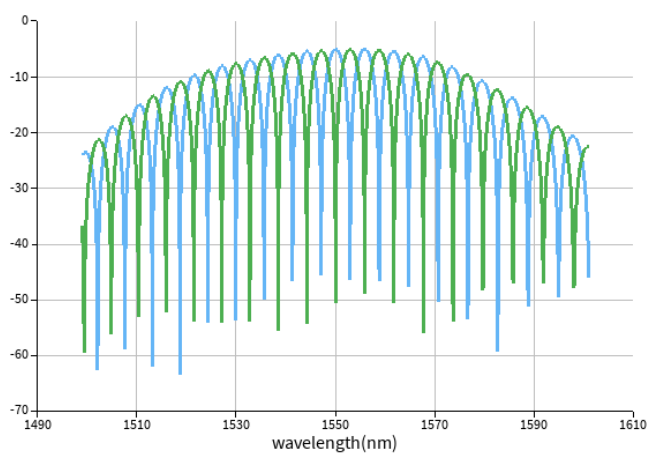
Layout Design (KLayout): The MZI variations were physically implemented in KLayout. The ΔL was achieved by introducing a "delay arm" in the upper branch, utilizing standardized waveguide bends and straight sections to ensure DRC (Design Rule Check) compliance.

PDK Components: Within **Lumerical INTERCONNECT**, the circuit was built using the library components corresponding to the foundry PDK. This includes:

- EBeam_gc_te1550: Grating coupler for fiber coupling in and out.
- EBeam_waveguide_Strip_TE_1550_nm_w=500_nm: waveguide characterized by the 500*220 nm cross-section.
- EBeam_bdc_te1550: 50/50 broadband splitter 1550nm, TE

5.2 Transmission Spectrum (Design #2)

The circuit simulation for Design #2 ($\Delta L = 100\mu m$) was executed using an Optical Network Analyzer (ONA). The resulting spectrum accounts for the combined effects of the waveguide dispersion and the port-to-port loss of the PDK components.



5.3 Verification of FSR

A critical step in the simulation is verifying that the PDK-modeled group index (n_g) aligns with our theoretical expectations. The distance between two adjacent peaks ($\Delta\lambda_{FSR}$) was measured directly from the INTERCONNECT ONA results. For the $100\ \mu m$ delay arm, the measured FSR is ~5.66 nm.

5.4 Theoretical Comparison:

Using the group index extraction formula:

$$n_g = \frac{\lambda^2}{\Delta\lambda_{FSR}\Delta L}$$

Substituting the simulation values ($\lambda = 1550$ nm, $\Delta\lambda_{FSR} = 5.66$ nm, and $\Delta L = 100\ \mu m$):

$$n_g = \frac{(1550 \times 10^3)^2}{5.66 \times 100} \approx 4.24$$

The simulated FSR is close to the predicted value from the theoretical model ($n_g \approx 4.1$). This confirms that the layout designed in KLayout and the compact models provided in the PDK are consistent. This provides high confidence that the fabricated devices will yield a measurable FSR suitable for n_g extraction.

6. Conclusion

This report has successfully demonstrated the theoretical derivation, design strategy, and simulation-based verification of unbalanced MZIs for group index extraction. Through Finite Difference Eigenmode (FDE) simulations, it was determined that the 500 nm x 220 nm silicon strip waveguide exhibits a high group index ($n_g \approx 4.059$) relative to its effective index ($n_{eff} \approx 2.44$), highlighting the significant role of structural dispersion in high-index-contrast SOI systems.

The implementation of a multi-device strategy—utilizing four different path length differences ($\Delta L = 50, 100, 150, 200\ \mu m$) - provides a robust framework for experimental validation. This approach allows for statistical averaging and ensures that a clear interference pattern is captured regardless of potential fabrication tolerances or narrow spectrometer spans. Circuit-level simulations confirmed the consistency of the design, with $100\ \mu m$ imbalance producing an FSR of 5.66 nm, which translates to an extracted n_g of 4.24. The close agreement between the theoretical models and the PDK-based simulations provides high confidence in the design's accuracy. Ultimately, these MZIs serve as a vital diagnostic tool for calibrating simulation models with fabricated results, enabling the precise timing and phase control required for complex silicon photonic systems.

References

[1] Chrostowski, Lukas, and Michael Hochberg. Silicon photonics design: from devices to systems. Cambridge University Press, 2015.